

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

MICROCONTROLLERS AND INTERFACING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) Assume suitable data wherever necessary.

1. Describe the listed instructions with examples and explain them in detail in terms of 8051 addressing modes.
 - (a) `MOVC A, @A + DPTR`
 - (b) `CPL A`
 - (c) `DA A`
 - (d) `CJNE A, #05h, L1`
 - (e) `RR A` 5(CO1)
2. Infer with diagram and explain pin and port configurations how an 8051 microcontroller can access a 4 KB external RAM. 5(CO2)
3. Recognize the flag structure of 8051 microcontroller with a neat diagram and implement a program to arrange 20 bytes of data present in data memory 30h onwards in ascending order. 10(CO2,3)
4. Structuring the use of timer, write an assembly language program, demonstrate serial transmission of a string "HELLO EC STUDENTS" at 4800 baud rate. Provide with details of calculation in regards to baud rate and mode selection of timer. Also illustrate what is the maximum baud rate generation possible with 8051. How one can double the baud rate with the use of a special register along with its format ? 10(CO3,4)

5. Implement and draw a circuit comprising of 8 LEDS (common Cathode configuration) connected at PORT 0 and a single switch is connected at PORT1. Write a program to display the amount of times switch is pressed across the LEDS. 10(CO3,4)
6. Describe ARM Controllers Privileged and Unprivileged Modes of operations with a flow diagram in detail. 5(CO1)
7. Describe ARM7TDMI with LPC2148 microcontroller architecture with diagram. 5(CO2)
8. Create a FIFO (First In First Out) algorithm for 4 tasks considering the input data given below in the table. Also, Explain what you understand by threads and how they differ from tasks.

Task No.	Arrival Time	Execution Time
1	1	3
2	7	5
3	10	9
4	15	2

10(CO5)

